CO3 Assignment — Divide and Conquer

CSA06 — Design and Analysis of Algorithms

Assignment	Assignment 3 – CO3
Student A	To be completed by the group
Student B	To be completed by the group
Total marks in supplied PDF	50
Deadline	Confirm current class/LMS deadline

Starter solution package. The supplied assignment PDF describes seven questions and requires genuine pair-work contributions. Q7's specific Day-4 challenge statement was not included in the uploaded 8-page brief, so this package provides independent-design templates instead of inventing the challenge.

1. Source-grounded assignment overview

The supplied brief identifies pair work for two students, says both must participate in every question, and warns that each student must be able to explain the complete assignment during verification. The PDF lists total marks as 50, although the class post says 100 points; confirm the grading total with the instructor/LMS.

Q1 — Max/Min and Binary Search

Max/Min uses recursive halves and combines the two local minima/maxima. For n=2^k, the tournament-style method uses C(n)=2C(n/2)+2 with C(2)=1, yielding 3n/2−2 comparisons. Binary search requires sorted data and halves the candidate interval each iteration: T(n)=T(n/2)+Θ(1)=Θ(log n).

Q2 — Merge Sort and Quick Sort

Merge Sort recurrence T(n)=2T(n/2)+Θ(n)=Θ(n log n), with Θ(n) auxiliary memory. Quick Sort has expected Θ(n log n) with randomized pivots, but deterministic extreme partitions can lead to Θ(n²). Record partition sizes and recursion depth in experiments; the included implementation emits comparison and partition statistics.

Q3 — Karatsuba Multiplication

For x=aB+b and y=cB+d, compute z2=ac, z0=bd, and z1=(a+b)(c+d)−z2−z0. Then xy=z2B²+z1B+z0. The recurrence is T(n)=3T(n/2)+Θ(n), so T(n)=Θ(n^log■3)≈Θ(n^1.585). Conventional multiplication may win for small inputs because recursion and slicing overhead dominate.

Q4 — Strassen Matrix Multiplication

Strassen reduces eight block multiplications to seven and uses additions/subtractions to reconstruct four result blocks. T(n)=7T(n/2)+Θ(n²)=Θ(n^log■7)≈Θ(n^2.807). The implementation pads non-power-of-two square matrices to the next power of two and uses a conventional multiplication cutoff. Padding and temporary arrays add costs.

Q5 — Closest Pair of Points

The brute-force baseline performs n(n−1)/2 distance calculations. Divide-and-conquer sorts points, solves each half, then checks a narrow y-sorted strip around the dividing line. The geometric packing argument limits relevant strip comparisons to a constant number per point; the standard recurrence is T(n)=2T(n/2)+Θ(n)=Θ(n log n). The included implementation is checked against brute force on sample data.

Q6 — Advanced Divide and Conquer

Median of Medians groups values in fives, selects a deterministic pivot, and gives worst-case Θ(n) selection time. QuickSelect is an expected-linear baseline but can degrade to Θ(n²) under poor pivots. Meet-in-the-Middle for subset sum enumerates roughly 2^(n/2) sums per half, combining by search; time and memory are exponential but substantially better than enumerating all 2^n subsets for suitable n.

Q7 — Day-4 Design Challenge

The uploaded assignment brief specifies a process for Q7 (independent proposals, comparison, joint design, recurrence, testing and cross-defence), but it does not state the actual Day-4 problem. The package deliberately leaves Student A and Student B proposal templates and the joint design template. Obtain the actual prompt, have both students independently complete their proposals before discussing, then implement and analyze the agreed design.

Experiments and evidence

Run Supporting_Files/test_all.py and Supporting_Files/run_experiments.py. Test results and a CSV timing log are included. Timing values are machine-dependent. No screenshots are fabricated: capture genuine terminal and source screenshots in the Output_Screenshots folders after running the code.

Final submission checklist

1) Fill in both students' names/registration numbers. 2) Complete actual Q7 challenge after obtaining its statement. 3) Each student writes their own Q7 proposal before discussion. 4) Capture genuine screenshots for all required modules. 5) Fill the contribution record based on actual work. 6) Re-run tests on your environment. 7) Confirm deadline and marks with LMS/faculty because the class post and PDF differ.